VISIONS CLASSES

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MAHARASHTRA STATE BOARD OF SECONDARY & HIGHER SECONDARY EDUCATION

SECONDARY SCHOOL CERTIFICATE (SSC) EXAMINATION

Mathematics (Part – I) | Standard IX

Chapter 3 : Polynomials — Practice Set 3.1

Total Marks : **15** Time : **45 Minutes**

General Instructions :

1. All questions are **compulsory**.
2. Use of calculator is **not allowed**.
3. Show all steps clearly. Figures to the right indicate marks.
4. For Q.3, attempt any **2 out of 3**.

Q. 1 Choose the Correct Alternative (MCQ) — 4 Marks (1 each)

(1) Which of the following is **NOT** a polynomial?

(A) $x^2 + 7x + 9$ (B) 10

(C) $y + 1/y$ (D) 5p

(2) What is the degree of the polynomial $7y - y^3 + y^5$?

(A) 1 (B) 3

(C) 5 (D) 7

(3) The coefficient form of the polynomial $x^3 - 2$ is :

(A) (1, 2) (B) (1, 0, 0, −2)

(C) (1, −2) (D) (1, 0, −2)

(4) Which of the following is a **quadratic** polynomial?

(A) 5p (B) $3r^3$

(C) $2x^2 + 3x + 1$ (D) $m^3 + 7m^2 + 5$

Q. 2 Complete the Following Activity **4 Marks**

(A) Write the polynomial $p(m) = 3m^5 - 7m + 5m^3 + 2$ in the required forms : *[2 Marks]*

Standard form : _____

Index form : $3m^5 +$ _____ $m^4 + 5m^3 +$ _____ $m^2 - 7m + 2$

Coefficient form : (_____ , _____ , _____ , _____ , _____ , _____)

Degree of the polynomial : _____

(B) Write the appropriate polynomials in the correct boxes : *[2 Marks]*

Given : $x + 7$, x^2, $x^3 + x^2 + x + 5$, $2x^2 + 5x + 10$, $x^3 + 9$, $3x^2 + 5x$

Quadratic polynomial : _____

Cubic polynomial : _____

Linear polynomial : _____

Monomial : _____

Binomial : _____

Trinomial : _____

Q. 3 Solve the Following (Any 2 out of 3) **4 Marks (2 each)**

(1) State whether the given algebraic expressions are polynomials. Justify your answer.

(i) $y + 1/y$

(ii) $2 - 5\sqrt{x}$

(iii) $x^2 + 7x + 9$

(iv) $2m^{-2} + 7m - 5$

(2) Write the following polynomials in coefficient form :

(i) $x^3 - 2$

(ii) $5y$

(iii) $2m^4 - 3m^2 + 7$

(3) Write the polynomials in index form (standard form) from the given coefficient forms :

(i) (1, 2, 3)

(ii) (5, 0, 0, 0, −1)

(iii) (−2, 2, −2, 2)

Q. 4 Solve the Following **3 Marks**

Write the polynomial in x using the given information :

(i) Monomial with degree 7 [1 Mark]

(ii) Binomial with degree 35 [1 Mark]

(iii) Trinomial with degree 8 [1 Mark]

Marks Distribution **For Reference**

Q. No.	Type	Topic	Marks
Q. 1	MCQ (4 × 1)	Polynomials, Degree, Coefficient form, Types	4
Q. 2	Activity (A + B)	Standard / Index / Coefficient form & Classification	4
Q. 3	Short Answer (Any 2 × 2)	Identify polynomials, Coefficient form, Index form	4
Q. 4	Long Answer	Writing polynomials — Monomial, Binomial, Trinomial	3
Total			**15**

*** Best of Luck *** | Be Honest, Work Hard

Visions Classes | Learn Today, Lead Tomorrow | Class 9 Mathematics (Part–I) — Chapter 3 : Polynomials